



Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 17

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Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	
Title of Assignment:	Write a Java program to demonstrate the use of Swing and Applets
Problem Statement & Learning Objective	Problem Statement Write a Java program to demonstrate the use of Swing and Applets Learning Objectives To develop graphical user interfaces using Swing components and Applets.
Theory	Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.
Program Code	Main1: (swing) <pre>import javax.swing.*; public class Main1 { public static void main (String[] args) { JFrame frame = new JFrame("Swing Demo"); JLabel label = new JLabel("Welcome to Java swing!", JLabel.CENTER); frame.add(label); frame.setSize(350, 200); frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); frame.setVisible(true); } }</pre>

```

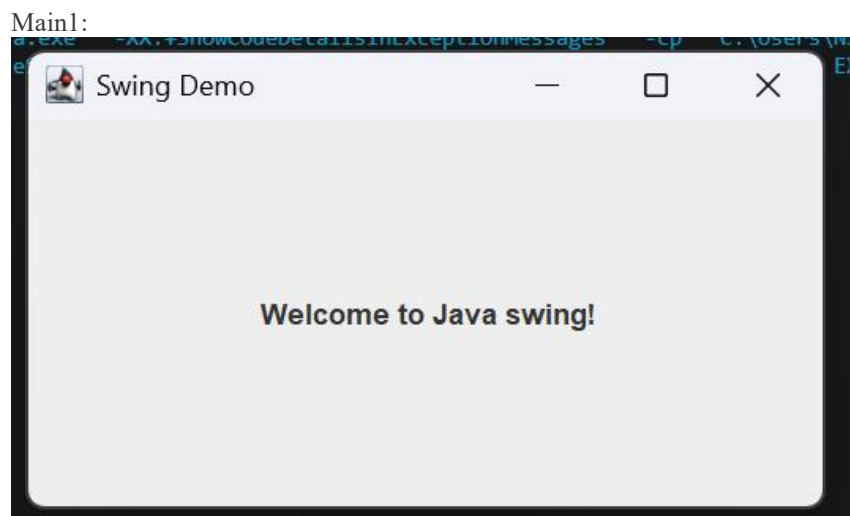
}
Main2: (applet)
import java.applet.*;
import java.awt.*;

public class Main2 extends Applet {
    public void paint (Graphics g) {
        g.drawString("Welcome to Java Applet", 50, 50);
    }
}
=====
AppletDemo.html:

<!DOCTYPE html>
<html>
<head>
    <title>Applet Demo</title>
</head>
<body>
    <applet code="Main2.class" width="300" height="150">
        Your browser does not support Java Applets.
    </applet>
</body>
</html>

```

Output Screenshots



Main2:
No output, Applet is depreciated.



**Exercise Qns
and Code**

1. Create a simple **Student Registration Form GUI** using Swing.

```
import javax.swing.*;

public class Exp1 {
    public static void main (String[] args) {
        JFrame frame = new JFrame("Student Registration Form");
        JLabel nameLabel = new JLabel("Name:");
        JLabel rollLabel = new JLabel("Roll No:");
        JLabel courseLabel = new JLabel("Course:");
        JTextField nameField = new JTextField();
        JTextField rollField = new JTextField();
        JTextField courseField = new JTextField();
        JButton submitButton = new JButton("Submit");
        nameLabel.setBounds(50, 50, 100, 30);
        nameField.setBounds(150, 50, 200, 30);
        rollLabel.setBounds(50, 100, 100, 30);
        rollField.setBounds(150, 100, 200, 30);
        courseLabel.setBounds(50, 150, 100, 30);
        courseField.setBounds(150, 150, 200, 30);
        submitButton.setBounds(150, 210, 100, 30);
        frame.add(nameLabel);
        frame.add(nameField);
        frame.add(rollLabel);
        frame.add(rollField);
        frame.add(courseLabel);
        frame.add(courseField);
        frame.add(submitButton);
```

```
        frame.setSize(420, 320);
        frame.setLayout(null);
        frame.setVisible(true);
```

```
        submitButton.addActionListener(e -> {
            JOptionPane.showMessageDialog(
                frame,
                "Registration Successful!\nName: " +
nameField.getText()
                + "\nRoll No: " + rollField.getText()
                + "\nCourse: " + courseField.getText()
            );
        });
    }
}
```

2. Create a simple **Employee Registration Form** using **Java Swing GUI** that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box

```
import javax.swing.*;

public class Exp2 {
    public static void main (String[] args) {
        JFrame frame = new JFrame("Employee Registration Form");
```

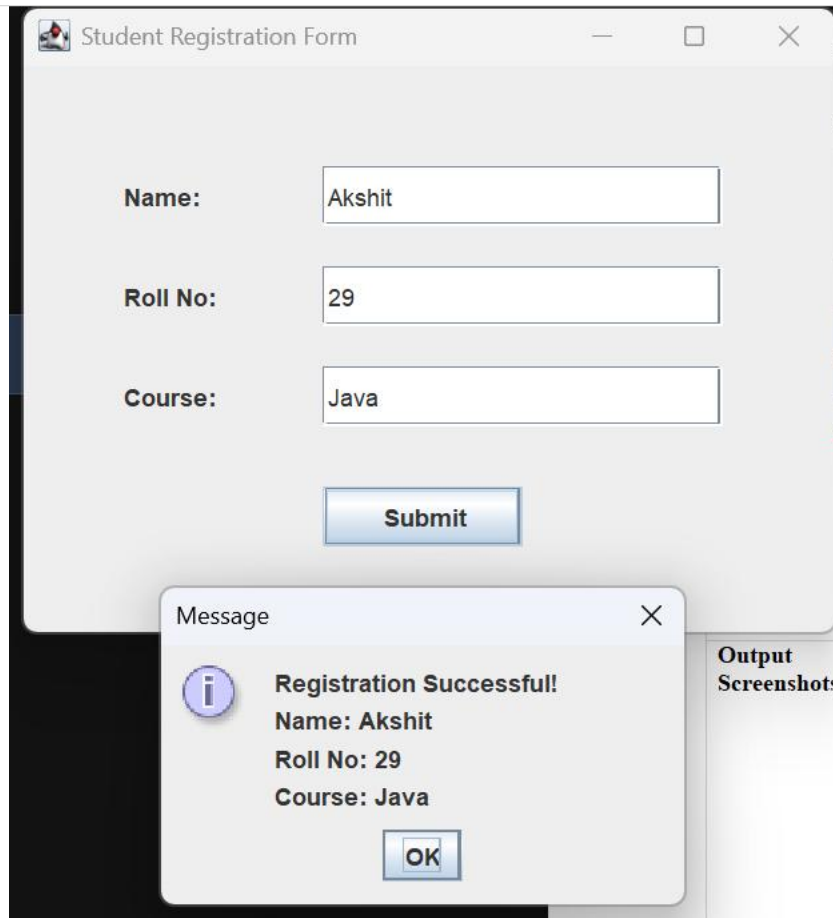
```

JLabel idLabel = new JLabel("Employee ID:");
JLabel nameLabel = new JLabel("Name:");
JLabel deptLabel = new JLabel("Department:");
JLabel salaryLabel = new JLabel("Salary:");
JTextField idField = new JTextField();
JTextField nameField = new JTextField();
JTextField deptField = new JTextField();
JTextField salaryField = new JTextField();
JButton submitButton = new JButton("Submit");
idLabel.setBounds(40, 40, 100, 30);
idField.setBounds(150, 40, 200, 30);
nameLabel.setBounds(40, 90, 100, 30);
nameField.setBounds(150, 90, 200, 30);
deptLabel.setBounds(40, 140, 100, 30);
deptField.setBounds(150, 140, 200, 30);
salaryLabel.setBounds(40, 190, 100, 30);
salaryField.setBounds(150, 190, 200, 30);
submitButton.setBounds(150, 240, 100, 30);
frame.add(idLabel);
frame.add(idField);
frame.add(nameLabel);
frame.add(nameField);
frame.add(deptLabel);
frame.add(deptField);
frame.add(salaryLabel);
frame.add(salaryField);
frame.add(submitButton);
frame.setSize(420, 350);
frame.setLayout(null);
frame.setVisible(true);
submitButton.addActionListener(e -> {
    JOptionPane.showMessageDialog(
        frame,
        "Employee Details\n"
        + "Employee ID: " + idField.getText()
        + "\nName: " + nameField.getText()
        + "\nDepartment: " + deptField.getText()
        + "\nSalary: " + salaryField.getText()
    );
});
}

```

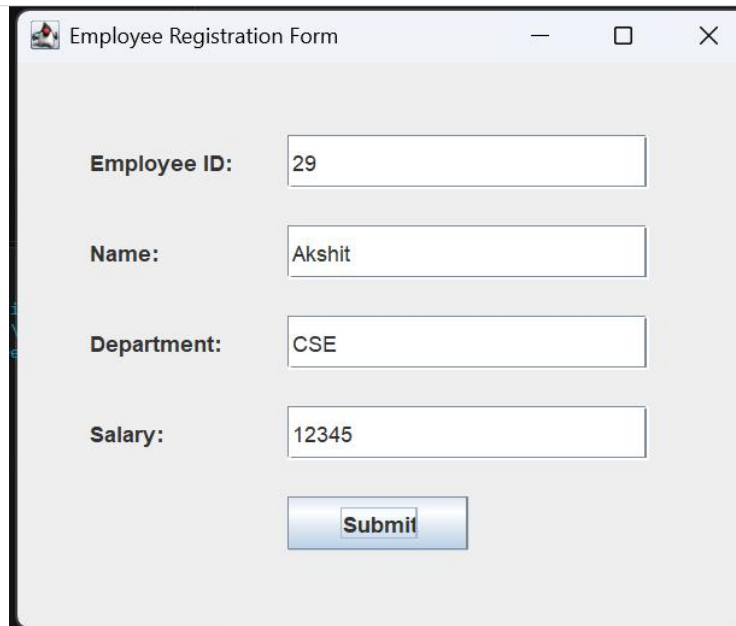
**Output
Screenshots**

Exp1:



Output
Screenshots

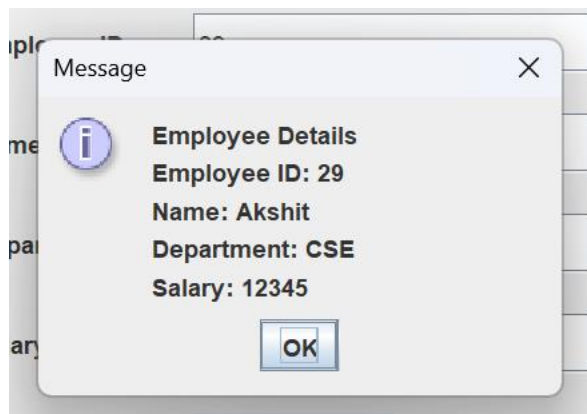
Exp2:



A screenshot of a Java Swing window titled "Employee Registration Form". The window has a standard title bar with minimize, maximize, and close buttons. Inside the window, there are four text input fields arranged vertically. The first field is labeled "Employee ID:" and contains the value "29". The second field is labeled "Name:" and contains the value "Akshit". The third field is labeled "Department:" and contains the value "CSE". The fourth field is labeled "Salary:" and contains the value "12345". Below these fields is a "Submit" button.

Employee ID:	29
Name:	Akshit
Department:	CSE
Salary:	12345

Submit



Conclusion

This experiment focuses on constructing Graphical User Interfaces using Java Swing components. By arranging containers and widgets like JFrame and JTextField, it details component layout management and cross-platform desktop UI development while contrasting modern Swing applications with legacy browser-based applets.